Synthetic Evaluation Benchmark. Faced with a shortage of human-generated data and observing that LLMs are approaching the quality of human-level annotations (He et al., 2023; Lee et al., 2023a), there has been a surge in research drawing inspiration from this development. Consequently, numerous studies have started utilizing LLMs to augment or synthesize large-scale dialogue benchmarks for assessing responses in everyday social interactions (Kim et al., 2023), examining responses in multi-modal environment (Feng et al., 2023), and evaluating responses that align with specific persona (Jandaghi et al., 2023). We leverage LLMs to create data but ensure its high quality with human verification and editing.

3 Generative Pipeline for LoCoMo

An overview of our generative pipeline for LoCoMo is shown in Figure 3. We create two virtual agents, named $\mathcal{L}_1$ and $\mathcal{L}_2$, each initialized with a LLM $\mathcal{M}$ (*i.e.,* `gpt-3.5-turbo`). To start, unique persona statements p are assigned to each agent $\mathcal{L}_i$, ensuring the integration of distinct personalities into their dialogues (§3.1). To mirror real-life experiences, we create a temporal event graph $\mathcal{G}$ for each agent, which illustrates a realistic sequence of life events (§3.2). The LLM agent architecture (Park et al., 2023) is utilized for each agent $\mathcal{L}_i$, enabling them to effectively memorize and reflect conversation history into ongoing dialogues (§3.3). Further, each agent $\mathcal{L}_i$ can share coherent images, thereby enhancing the multi-modal dialogue aspect. Finally, human annotators are tasked with manually filtering and refining the generated data (§3.4).

3.1 Persona

We select an initial persona statement p_c from the MSC dataset (Xu et al., 2022), encompassing 4 to 5 sentences, and employ `gpt-3.5-turbo` as $\mathcal{M}$ to expand these into full persona statement p (See examples and prompt details in Appendix A.1). The generated statements typically include details about one or more of the following elements (Gao et al., 2023a): objectives, past experiences, daily habits, and interpersonal relationships, as well as name, age, and gender of the individual.

3.2 Temporal Event Graph

To utilize the real-life experiences of each agent in the conversation, we construct a temporal event graph, labeled as $\mathcal{G}$, for each agent. This graph $\mathcal{G}$, consisting of events e_i, is produced by applying the condition of $\mathcal{M}$ (i.e., `text-davinci-003`) on a designated persona p. Each event e_i is associated with a date of occurrence t_i. $\mathcal{G}$ includes causal connections $l = (e_i, e_j)$ that illustrate the causal relationships among events $e_i \in \mathcal{G}$ and reflect a natural succession of events in an individual's life. For each $\mathcal{G}$, we create up to 25 events, spread across a time frame of 6 to 12 months, in an iterative process that balances between inference time and the coherence of temporal and causal connections in the timeline. Initially, a small batch of $k = 3$ events is generated, which is then used iteratively as input prompt to create the subsequent batch of k events. See details in Appendix A.2.

3.3 Virtual Agent Architecture

Every agent $\mathcal{L}_i$ incorporates modules from generative agent architecture (Park et al., 2023). The agent has two functions: (1) *reflect & respond*; and (2) *image sharing & image reaction*. The agent is asked to primarily use the *reflect & respond* function while employing *image sharing & image reaction* function judiciously and appropriately within the context of the conversation.

Reflect & Respond. The fundamental process for each agent to *reflect and respond* involves the concept of short-term and long-term memory. During inference, agent $\mathcal{L}_i$ conditions its responses on both short and long-term memories, paralleling how humans remember recent conversations while also recalling distilled important experiences from long-term memory. Following each session k, each agent is asked to produce a summary w_k that is then stored in the short-term $\mathcal{H}_s$. This summary w_k is generated by conditioning $\mathcal{M}$ on both the most recent session conversation history h_k and the preceding summary $w_{k-1} \in \mathcal{H}_l$. For each turn j within session k, a single turn of the conversation h_{k_j} is transformed into an observation o_{k_j} and then stored in the long-term memory $\mathcal{H}_l$. Then, agent $\mathcal{L}_i$ generates a response in session $k+1$ on the date t^s_{k+1} by basing it on the latest summary w_k, reflections based on the retrieved relevant observations $o \in \mathcal{H}_s$, the ongoing conversation history in the current session h_{k+1} and persona statement p. Long-term temporal narratives are induced in the conversation by additionally conditioning the agent's response on the subset of events in $\mathcal{G}$ that occur between the last and current session i.e. $\{e \in \mathcal{G} \,|\, t^s_k < t^e_i < t^s_{k+1}\}$. See details in Appendix A.2.1.

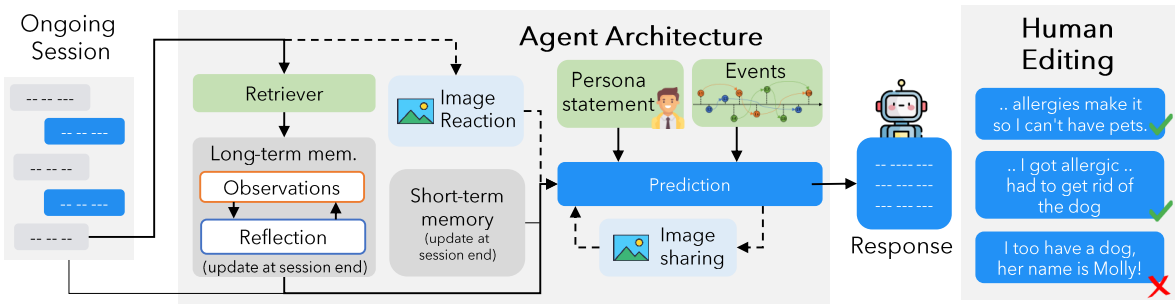

Figure 3: **Overview of the generative pipeline for LoCoMo.** Each LLM agent is assigned a distinct persona and a timeline of causally connected events in their file. The agent is equipped with a memory and reflection module to retrieve relevant history for dialog generation and is also enabled for image-sharing and image-reaction behaviors (left). The generated conversations are edited by human annotators to maintain long-range consistency (right).

Image Sharing & Image Reaction. The *image sharing & image reaction* functions are integrated to add a multi-modal dimension to the long-term dialogues.[2] The *image sharing* function is called when the agent decides to send an image. This process includes: (1) Generate a caption c for the intended image using $\mathcal{M}$; (2) Convert the caption c into relevant keywords w using $\mathcal{M}$; (3) Use the keywords k to find an image through web search $WEB(k)$[3]; (4) Share the chosen *image*. Conversely, the *image reaction* function is triggered upon receiving an image from another agent and entails: (1) Generate caption c for the received image[4]; (2) Generate a reaction for the received image in response using $\mathcal{M}$ (See Appendix A.2.1).

3.4 Human Verification & Editing

In the concluding phase, human annotators are tasked with (1) editing the dialogue to eliminate long-term inconsistencies, (2) removing or substituting irrelevant images, and (3) verifying and editing for alignment between event graphs and the content of the conversations. Overall, we observed that annotators edited nearly 15% of the dialog turns and removed or substituted approx. 19% images present in the LLM-generated dataset. See examples of some edits in Appendix A.3.

4 LoCoMo Evaluation Benchmark

Based on the dialogues generated in section 3, we introduce an evaluation benchmark (see Figure 2) composed of three tasks to assess the accuracy of *long-term memory*. See statistics of the dataset and evaluation benchmark in Table 5 in the Appendix.

[2]Image captions are also saved to long-term memory.
[3]https://pypi.org/project/icrawler/
[4]We use BLIP-2 (Li et al., 2023b) as the captioning model.

4.1 Question Answering Task

A conversational agent is expected to possess a *memory* to remember previous dialogues, reflecting it to create more engaging responses in future conversations. For a comprehensive assessment of this *memory*, we introduce a question-answering task divided into five distinct reasoning categories: (1) **Single-hop** questions require answers based on a single session; (2) **Multi-hop** questions require synthesizing information from multiple different sessions; (3) **Temporal reasoning** questions can be answered through temporal reasoning and capturing time-related data cues within the conversation; (4) **Open-domain knowledge** questions can be answered by integrating a speaker's provided information with external knowledge such as commonsense or world facts; (5) **Adversarial** questions are designed to trick the agent into providing wrong answers, with the expectation that the agent will correctly identify them as unanswerable.

For each category, we calculate the F1 score for exact matches, following the normalization of both the predicted and the actual ground truth answers. However, evaluating long-form answers with automated metrics often presents challenges (Xu et al., 2023). LLMs tend to produce paraphrased responses in varied formats, complicating exact match evaluation. To simplify evaluation in our task, we ensure that answers in our QA annotations are directly taken from the conversations as much as possible. We instruct the LLMs to replicate the exact wording in the conversation when feasible and employ the F1 partial match metric for evaluating the predictions. Each QA sample is also annotated with the turn IDs in the conversation logs that contain the answer. We report the accuracy of retrieving the correct context for RAG models.

4.2 Event Summarization Task

The conversation is generated based on a temporal event graph $\mathcal{G}$ which is constructed by conditioning an LLM on a persona statement p, reflecting the chronological sequence of events in an individual's life. A conversational agent is expected to not only comprehend the causal connections and the sequence of events in $\mathcal{G}$ but also to recount these events as required. To evaluate the agent's grasp of event dynamics, we introduce the event summarization task which challenges the agent to summarize the events within a designated timeframe and compares the agent's summary with events in $\mathcal{G}$. The events in LoCoMo are densely annotated lists of life events that are hard to summarize due to temporal and causal coreferences present in the dialogues, in contrast to existing summarization benchmarks of research papers (Li et al., 2023a), movie scripts (Chen et al., 2022), books (Kryściński et al., 2022), emails (Zhang et al., 2021b) etc.

Traditional metrics like BLEU (Papineni et al., 2002) and ROGUE (Lin, 2004) focus on lexical similarity between the reference and generated summaries, not meeting our needs as we emphasize factual accuracy in summarization. In this context, we employ FactScore (Min et al., 2023), a method that evaluates the factuality of generated text by decomposing both the reference and hypothesis into atomic facts. We adapt the metric to measure (1) *precision* of the summarized content by counting the number of atomic facts within the content that correspond with those in $\mathcal{G}$; (2) *recall* of the summarized content by determining how comprehensively the atomic facts of $\mathcal{G}$ are represented within the content. We present the F1 score, derived from the calculated precision and recall.

4.3 Multi-Modal Dialogue Generation Task

The conversations in our dataset are anchored to specific personas p and corresponding events $\mathcal{G}$ tailored to p. The topics in conversations evolve from events that were introduced in earlier dialogues, spanning weeks or months. This structure allows for an assessment of whether conversational agents can sustain a coherent persona and a continuous narrative over time. For example, if a speaker recently had an injury, the next conversations would likely focus on them recuperating, rather than engaging in adventurous activities. We assess such consistency by measuring how closely the predicted multi-modal dialogues align with the ground truth multi-modal dialogues in our dataset, quantifying this alignment through MMRelevance (Feng et al., 2023), in addition to other NLG metrics.

5 Experimental Setup

For the question-answering and event summarization tasks, we replace images in LoCoMo with their captions (Li et al., 2023b), and use state-of-art LLMs to reason over text-only dialogues interleaved with image captions. We use images directly for the multimodal dialog generation task only. See additional details in Appendix C.

Question Answering. We evaluate three types of models: (1) **Base** LLMs operating with constrained context lengths where earlier dialogues are omitted i.e., Mistral-7B (Jiang et al., 2023), LLama-70B-chat (Touvron et al., 2023), `gpt-3.5-turbo` [5], and `gpt-4-turbo` [6]; (2) **Long-context** LLMs with an extended context window i.e., `gpt-3.5-turbo-16k`; (3) **Retrieval-augmented Generation (RAG)** involves retrieving relevant context from a database of dialog history, observations (assertions about speakers; see §3.3, Figure 9), or session-level summaries (see §3.3, Figure 8). We employ DRAGON (Lin et al., 2023) as retriever and `gpt-3.5-turbo-16k` as reader.

Event Summarization. We present experiments using **Base** and **Long-context** setups from the question-answering task, but refrain from including RAG since summarization requires a comprehensive understanding of the entire dialogue, rather than just retrieving a specific portion. We implement incremental summarization i.e., iteratively create a summary of a preceding sessions and then use that summary as a basis to summarize the subsequent sessions (Chang et al., 2023).

Multi-modal Dialogue Generation. We generate 50 conversations using our automated pipeline (without human filtering; §3) for training data and train three versions of MiniGPT-5 (Zheng et al., 2023): (1) **Base** trains on prior dialogue turns only; (2) **+ summary** trains on prior dialogue turns and a global summary of the ongoing conversation; (3) **+ observation** trains on prior dialogue turns and observations retrieved from conversation history. Each run is initialized with a MiniGPT-5 checkpoint finetuned on MMDialog (Feng et al., 2023).

[5] https://platform.openai.com/docs/models/gpt-3-5

[6] https://platform.openai.com/docs/models/gpt-4-and-gpt-4-turbo